

Employment

Software Engineer

May 2025 – Present

RelationalAI, Logic Engine team

Remote

- Design and implement the imperative extension to RelationalAI's query language
 - Cut previously intractable enterprise queries to sub-hour runtimes
- Build the GQL-style graph pattern matching extension for the query language
- Maintain the database engine's Protocol Buffer-based entry point (~100k queries/day)
- Develop an event-based user-facing logging tool to diagnose slow queries

Graduate Researcher

Sep 2022 – Apr 2025

Northeastern University, Programming Research Lab

Boston, USA

- Designed a language to express optimization problems under uncertainty [1]
 - Implemented a novel binary decision diagram algorithm to compute optimal values
 - Beat a state-of-the-art probabilistic programming language in optimization benchmarks
- Co-led the first empirical study of online proof assistant communities [2]

R&D Intern

May 2024 – Dec 2024

RelationalAI, Logic Engine team

Remote

- Built a reverse compiler for Datalog to surface query optimizations for easier debugging
- Integrated GraphViz visualizations of query optimizations into the query language's profiler

REU Researcher

Jun 2021 – Jul 2021

University of Tennessee

Chattanooga, USA

- Published research in the theory of matrices in function spaces [3]
- Automated routine computation in Mathematica, saving weeks of manual effort

Education

Master of Science, Computer Science

Sep 2022 - May 2025

Northeastern University

Boston, USA

Bachelor of Science, Mathematics and Philosophy

Aug 2018 - May 2022

Carnegie Mellon University

Pittsburgh, USA

Publications

- [1] Minsung Cho, John Gouwar, and Steven Holtzen. "Scaling Optimization over Uncertainty via Compilation". In: *Proc. ACM Program. Lang.* 9.OOPSLA1 (2025). DOI: 10.1145/3720500.
- [2] Gwentyth Lincroft*, Minsung Cho*, et al. "Thirty-Three Years of Mathematicians and Software Engineers: A Case Study of Domain Expertise and Participation in Proof Assistant Ecosystems". In: *21st IEEE/ACM International Conference on Mining Software Repositories, MSR 2024*. ACM, 2024, pp. 1–13. DOI: 10.1145/3643991.3644908. *Joint first authors.
- [3] Minsung Cho, Seth Hoisington, Roger Nichols, and Brian Udall. "The Krein-von Neumann extension of a regular even order quasi-differential operator". In: *Opuscula Mathematica* 41.6 (2021), pp. 805–841. DOI: 10.7494/OpMath.2021.41.6.805.

Awards and Recognition

First Place, ACM Student Research Competition at PLDI

2023

Scholarship Recipient, Programming Languages Mentoring Workshop (PLMW) at PLDI

2023

University and College Honors, Carnegie Mellon University

2022

Skills

Programming languages: Julia, Python, Datalog, OCaml, Rust, Go, C/C++, Haskell, Standard ML

Tools: Unix, Git, Jira, Docker, Datadog, Observe, Snowflake